Instructions:

Create a team of up to 5 people for solving short assignments.

Task:

Open EViews and the workfile "VARS", click on the page named "pl". The page contains quarterly data on time series for the Poland economy from Q1-2000 to Q2-2026:

- Harmonised index of consumer prices ('cpi').
- GDP at constant 2020 prices ('gdp').
- Zloty real effective exchange rate ('reer').
- Zloty 3 months money market interest rate ('rs').
- Nominal Brent oil price ('oil').

1. Structural VAR: Identifying a Monetary-Policy Shock

a. Estimate the reduced-form VAR in four endogenous variables: 'reer', 'gdp', 'cpi', 'rs'. To ensure stationarity, include variables to the VAR in log-differences (multiply by 100), except the interest rate 'rs' (include 'rs' in levels). Include the log-diff of 'oil' and an intercept as exogenous variables. Estimate the VAR on 4 lags of endogenous variables. Use the sample from Q1-2005 to Q4-2024.

b. Check if the VAR from part (1) is stationary. Test the residuals for autocorrelation using the LM Test. Make a conclusion about using the VAR for policy analysis and forecasting.

c. Look at the residual's correlation matrix. Are the off-diagonal correlations (roughly) zero? Explain the consequences for impulse-response analysis.

d. Impose recursive restrictions on the contemporaneous relations among the endogenous variables. Provide an economic intuition for recursive ordering.

Hint: In the 'Impulse Definition' window choose 'Cholesky – dof adjusted' method and order the variables 'Exchange rate' → 'GDP' → 'Inflation' → 'Interest rate'.

Use bootstrapped standard errors for IRFs ('Kilian's unbiased') because the residuals are non-normal and the sample is short. Plot the IRF for 16 quarters. Estimate IRFs twice: non-accumulated and accumulated responses.

e. Read the non-accumulated impulse responses. Focus on the column of responses to the interest rate shock (monetary policy shock) and record what you see: sign and magnitude on impact, responses after one and two years, when the variable stabilizes.

f. Read the accumulated impulse responses. Explain the differences with part (e). Which responses are simpler to interpret and communicate?

g (optional). Re-estimate the IRFs with the opposite order of the variables and compare them to parts (e) and (f). What do you see?

h (optional). Using the results from part (f), calculate the peak responses of GDP and CPI to the 1 p.p. monetary policy shock. What do you think about the strength of the monetary policy transmission mechanism in Poland?

i (optional). Estimate the forecast error variance decomposition using the same order as in part (d). Which shocks do drive the dynamics of the endogenous variables? Are your results useful for policymakers?

Discussion questions – Part 1

1. Why can't we interpret a single reduced-form residual (say, the 'rs'-equation residual) as "the monetary-policy shock" directly?

2. In plain words, what does ordering the policy interest rate last assume about how the central bank behaves within a quarter?

3. Give one real-world reason the ordering *GDP → Inflation → Interest rate* might be wrong for some economies.

2. Forecasting with a VAR

Use the estimated reduced-form VAR from part (1.a) for this task.

a. Turn the VAR into a Model object.

b. Solve the baseline dynamic forecast. In the dialog set:
- Simulation type: Deterministic.
- Dynamics: Dynamic (uses forecasted lags to roll forward – a multi-step forecast).
- Solution sample: 2025q1 2026q1.
- Active scenario: Baseline.

Hint: EViews saves the forecasts with a _0 suffix (e.g. gdp_0).

c. Compare the baseline forecast with the actuals. Does the forecast track well early and drift later? Which variable is forecast worst?

Hint: Make graph from the Model object for endogenous variables over 2022–2026.

d. Simulate an alternative scenario – an aggressive rate-hike path. Compare with the baseline scenario. Do higher prices in the alternative scenario mean that rate hikes cause inflation?

Hint: Create Scenario 1 in the Model object. Solve the model for Scenario 1. Override the 'rs' variable – enter +0.25 for Q1-2025, +0.50 for Q2-2025, +0.75 for Q3-2025, +1.00 for Q4-2025 – Q2-2026 ('Delta'). Solve Scenario 1 again. Make Graph: compare Scenario 1 vs Baseline vs Actuals for all endogenous variables.

e (optional). Add forecast uncertainty (bounds) to the baseline. Do the actuals stay inside the bands?

Discussion questions – Part 2

1. Why does the dynamic forecast usually get worse the further ahead you go.

2. Which part of today's work would you trust to answer "what happens to inflation if the central bank raises rates?" – Part 1 or Part 2, and why?

3. Name one variable you would add to improve the forecast, and say whether it should be endogenous or exogenous.